TRALSem: A Robust Model for Textual Sentiment Analysis

Bo Yang, Jiayi Dang, Huai Liu, *Senior Member, IEEE*, and Zhi Jin, *Fellow, IEEE*

***Abstract*—Sentiment analysis has gained widespread applications across various domains due to its versatility and practicality. With the increasing availability of data and advancements in machine learning technologies, its utilization is expected to continue expanding. Deep learning (DL)-based sentiment analysis has demonstrated high accuracy and efficiency in numerous application areas, such as marketing, customer service, politics, healthcare, and finance, thereby highlighting its high potential. Despite recent progress, DL-based sentiment analysis methods still face significant challenges, particularly concerning the robustness of sentiment classification and scoring. To address these issues, our study introduces TRALSem, a novel text-centered sentiment analysis framework designed to tackle the unique challenges of sentiment classification and scoring, with a particular focus on enhancing the overall robustness of the model. We conducted extensive experiments on multilingual script datasets, including Chinese and English scripts, as well as the IMDB and SST datasets. The experimental results show that TRALSem significantly outperforms existing state-of-the-art sentiment analysis methods. In terms of sentiment classification, it achieves remarkable improvements, far surpassing the previous benchmarks. More importantly, TRALSem substantially enhances the model's robustness, enabling it to maintain stable performance even in the face of complex and noisy data. It also significantly reduces the model's sensitivity to noise data, effectively filtering out interference and providing more reliable and accurate sentiment analysis results. Moreover, it offers better interpretability to users, making the sentiment analysis process and outcomes more understandable and actionable.**

***Index Terms*—Deep learning (DL), pretrained model, robust, sentiment analysis.**

I. Introduction

SENTIMENT analysis is an important research direction in the field of natural language processing [2], [3], [4], focusing on extracting sentiment, attitude, and emotion information from texts. Sentiment analysis is extensively utilized across numerous domains, including social media monitoring [5], product recommendation [6], opinion mining [7], customer service [8], [9], among others. Sentiment analysis can help organizations better understand consumer needs and feedback, thereby optimizing products and services.

Presently, sentiment analysis methodologies can be classified into three primary categories: those based on traditional sentiment dictionary approaches, traditional machine learning (ML)-based methods, and deep learning (DL)-based methods. Sentiment dictionary-based traditional methods depend on a preconstructed sentiment lexicon or dictionary, which encompasses a collection of words each associated with a particular sentiment polarity—positive, negative, or neutral. The sentiment score for a given text is derived by summing the sentiment scores of the words that comprise it. Typical examples of such traditional methods include the TextBlob library [10] and the SentiWordNet lexicon [11].

ML-based techniques have been very successful in many fields, such as natural language translation [12], [13], [14], healthcare [15], [16], [17], [18], facial recognition [19], [20], cloud computing [21], etc. ML-based techniques harness ML algorithms to discern the correlation between textual features and sentiment classifications from annotated training datasets. Frequently employed ML algorithms for sentiment analysis include Naive Bayes, support vector machines (SVM), and random forests. ML-based approaches necessitate feature engineering, a process that involves selecting and extracting pertinent features from the text, such as n-grams, part-of-speech tags, and syntactic dependencies. Uddin et al. [22] created Sentisead, a monitoring system that employs polar labels and word clusters as its primary features. Huq et al. [23] utilized the K-nearest neighbor (KNN) and SVM algorithms to determine the emotional orientation of Twitter content.

DL-based approaches leverage deep neural networks to automatically extract hierarchical representations of text features and sentiment labels from extensive unlabeled data. DL-based approaches have demonstrated high levels of performance across numerous natural language processing tasks, including sentiment analysis. Notable examples of DL-based models are convolutional neural networks (CNN), recurrent neural networks (RNN), transformer-based models such as BERT and GPT, and so on. For example, Ouyang et al. [24] introduced a structure based on Google's Word2vec and employed it to construct a tailored CNN framework for sentiment analysis. Liu et al. [25] introduced a Roberta-based model that demonstrated improved efficacy by incorporating dynamic masking

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Bo Yang and Jiayi Dang are with the School of Information Science and Technology, Beijing Forestry University, Beijing 100083, China (e-mail: yangbo@bjfu.edu.cn; dang_jiayi@163.com).

Huai Liu is with the Faculty of Department of Computing Technologies, Swinburne University of Technology, Hawthorn, VIC 3122, Australia (e-mail: hliu@swin.edu.au).

Zhi Jin is with the Faculty of School of Computer Science, Peking University, Beijing 100871, China (e-mail: zhijin@pku.edu.cn).

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techniques. Zhang et al. [26] developed the ERNIE2.0 model by merging the strategy of continuous multitask learning and adding a range of tasks with varying levels of granularity. Furthermore, Onan [27] combined LSTM with the GloVe word embedding scheme to achieve the highest prediction performance on the MOOC review dataset, reaching a classification accuracy of 95.80%. Additionally, Onan [27] fused long short-term memory (LSTM) networks with the GloVe word embedding method to attain better performance.

However, the robustness of DL-based sentiment analysis models is one major challenge that needs to be addressed. The following lists major reasons why DL-based sentiment analysis models may not be robust. First, DL models can be susceptible to adversarial training. In adversarial training, small perturbations are added to the input text to alter the model's predictions without significantly changing the meaning of the text. Adversarial training can be used to deceive the model, thereby compromising its accuracy. Second, DL models can be highly sensitive to label noise, which arises when the training data contains incorrect labels or noisy examples. Label noise can lead to overfitting and poor generalization performance. Last but not least, although existing sentiment analysis models can perform well on single-type datasets, they require further improvements when applied to multilingual and multitype datasets. To address the aforementioned limitations, our model primarily aims to address the challenges of multilingual text and multitype classification, while reducing the impact of label noise on the dataset.

Strengthening the robustness of DL models [28], [29] can enhance the efficacy of sentiment analysis. By enhancing the robustness of DL models, we can bolster their precision and generalization capabilities, which can ultimately result in more efficient sentiment analysis. There are different ways to enhance the robustness of DL models, thereby enhancing the efficiency of sentiment analysis:

Improved Accuracy: Robust DL models are less susceptible to noise and adversarial training, which can lead to more accurate sentiment analysis results. This can reduce the need for manual review and correction of sentiment analysis outputs, which can save time and resources.

Better Generalization: Robust DL models can generalize better or even to new domains and data types, which can reduce the need for retraining or fine-tuning the model for each new application. This can save time and resources, thus making sentiment analysis more scalable.

Increased Automation: Robust DL models can automate more aspects of sentiment analysis, such as identifying sentiment expressions and classifying sentiment polarity. This can reduce the need for human annotation and intervention, which can save time and resources and make sentiment analysis more efficient.

In summary, enhancing the robustness of DL models can have significant benefits for sentiment analysis. In this article, we propose a robust model, namely TRALSem, which uses a transformer structure with three optimization strategies from the above three aspects. They are the additions of random seed (RS), adversarial training methods (ATM), and a linear layer (LL).

The article's contributions are three-fold, outlined as follows.

1) The sentiment analysis model introduced in this study, TRALSem, is designed to be robust, leveraging the superior performance of a pretrained model while implementing a range of optimization strategies. These strategies are instrumental in enhancing the robustness of the model.
2) TRALSem, along with a spectrum of standard methods, has been employed across English and Chinese datasets. On both linguistic datasets, TRALSem demonstrated superior performance compared with the other methods.
3) The proposed TRALSem model has been implemented and assessed across four datasets, as well as against 15 conventional methods. The experimental outcomes indicate that our approach outperforms the state-of-the-art techniques.

In a preliminary study [1], we have proposed a lightweight text-based sentiment analysis model. In this article, we not only refine this method further but also provide an in-depth investigation and discussion of the impact of the three optimization strategies (RS, ATM, and LL) on the robustness of TRALSem. In addition, we upgrade the method by adding four strategies, namely the fast gradient method (FGM) [30], projected gradient descent (PGD) [31], free large-batch (FreeLB) [32] and you only propagate once (YOPO) [33], to enhance the robustness, and experiments have been conducted to evaluate the influences of these four strategies on the robustness. A comparative experiment on the performances of different combinations of LL and softmax layer was also added.

The remainder of this article is structured as follows. Section II presents the related knowledge. TRALSem is described in Section III. Section IV presents the experiments for evaluating TRALSem, the results of which are reported in Section V. Section VI presents the related work. Finally, we conclude this article in Section VII.

II. Background

A. Sentiment Analysis

Sentiment analysis, as a crucial natural language processing technique, involves analyzing text to determine the sentiment expressed by the author. This can involve identifying positive or negative emotions such as happiness, sadness, anger, or frustration [34]. Sentiment analysis has been widely used in diverse applications such as social media surveillance, customer support, market analysis, and more cite DBLP:journals/ftir/PangL07. An application of sentiment analysis includes offering valuable perspectives on public sentiments with respect to specific subjects or products. By analyzing large volumes of social media posts or online reviews, companies can gain a better understanding of how customers perceive their brand and identify areas where improvements can be made citeDBLP:conf/aaaiss/MishneG06. Additionally, sentiment analysis can be used to monitor social media channels in real time, allowing companies to quickly respond to negative feedback or issues before they escalate cite DBLP:journals/corr/abs-1801-07883.

The application of sentiment analysis has grown rapidly in recent years due to its ability to provide valuable insights into consumer behavior and attitudes towards brands and products [35]. Some common applications include social media

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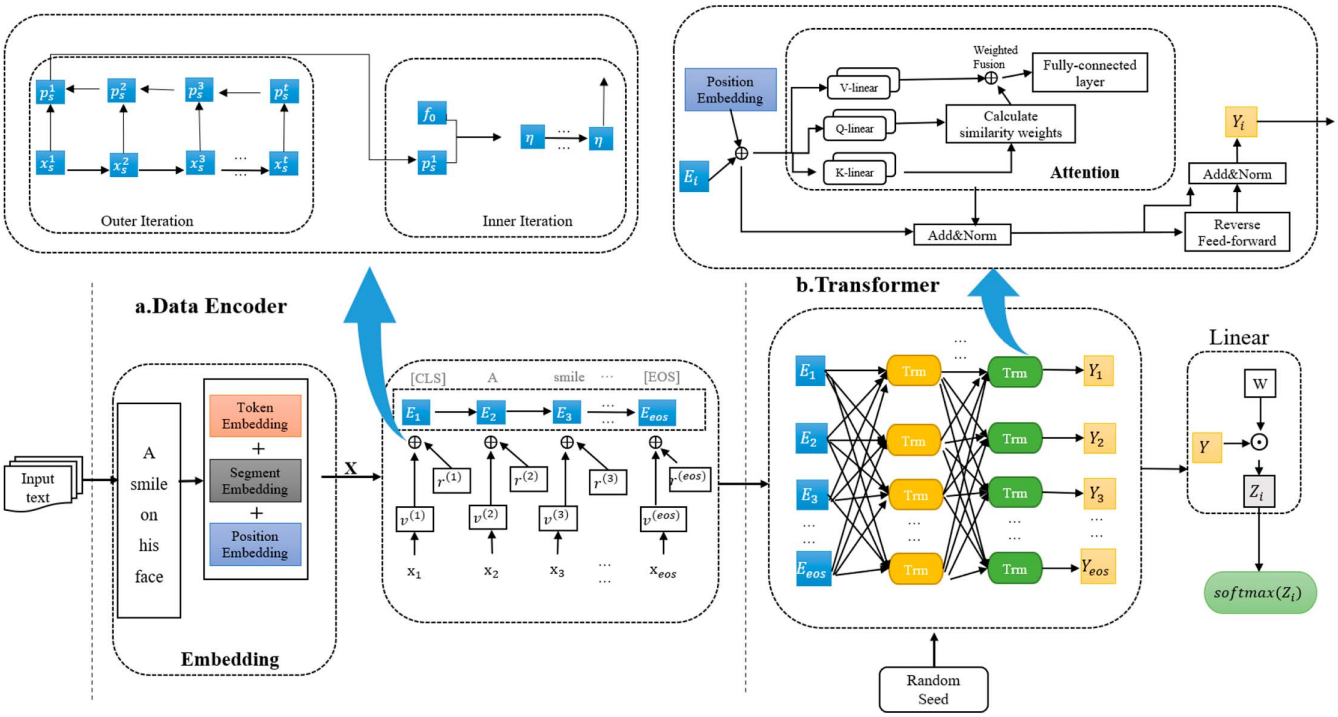


Fig. 1. TRALSem model.

monitoring, customer service, market research, and political analysis. For example, companies can use sentiment analysis to monitor social media posts and online reviews to understand how customers feel about their products and services [36], while political campaigns can use sentiment analysis to analyze public opinion on key issues and candidates.

B. Robust DL Model

The robust DL model, a type of deep neural network architecture, is engineered to be more resilient to input data variations compared with traditional DL architectures [37]. This enhanced robustness is attained by integrating additional layers or modifying existing ones within the network to boost its capacity to handle noise or outliers in the input data. The robust DL model has demonstrated high performance across a broad spectrum of tasks, including image classification [38], object detection [39], and speech recognition [40].

The applications of the robust DL model are diverse, spanning from computer vision and natural language processing to bioinformatics and finance [41]. In computer vision, the robust DL model has been applied for object detection and segmentation in images and videos. For instance, in the context of autonomous driving, accurate object detection in video feeds is crucial, and robust DL models have enabled significant progress in this area [42]. In natural language processing, it has been utilized for sentiment analysis [43] and named entity recognition [44]. In bioinformatics, the robust DL model has played a role in protein classification and drug discovery. In finance, it has been employed for stock price prediction and risk management [45], where accurately predicting market trends and assessing risks are of utmost importance.

C. Pretrained Model

A pretrained model is a DL model that has already been trained on a large dataset before being made available for use by others. A pre-training process typically involves fine-tuning a basic model architecture with additional layers or changes to existing layers to improve performance on a specific task. Once the pretraining process is complete, the resulting model can be reused and adapted for many different tasks without requiring additional training from scratch. Utilizing this method can drastically reduce time and computational requirements compared with developing a new model from scratch for each emerging task [46]. The application of pretrained models has become increasingly popular in recent years due to their ability to achieve high performance on a wide range of tasks with minimal additional training [47]. Some common applications of pretrained models include computer vision, natural language processing, and speech recognition. For example, pretrained models such as VGG [48], ResNet [49], and Inception [50] have been widely used for image classification and object detection tasks. In the domain of natural language processing, pretrained models such as BERT [47], GPT-3 [51], and Roberta [25] have been employed for tasks such as sentiment analysis [43], question answering [52], and language translation [53].

III. Methodology

A. TRALSem Model

The architecture of the TRALSem is illustrated in Fig. 1. The model consists of six components: the input layer, data encoding, adversarial training, a transformer with an RS, and a linear output layer. Initially, TRALSem performs word segmentation

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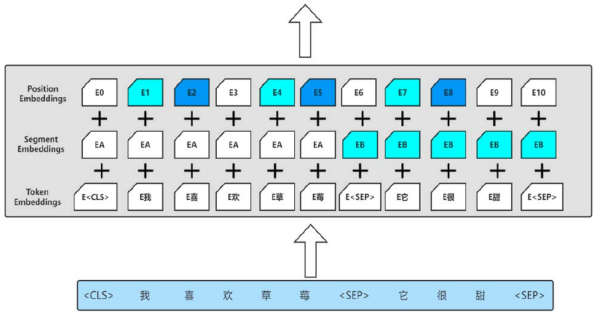


Fig. 2. Details of embeddings.

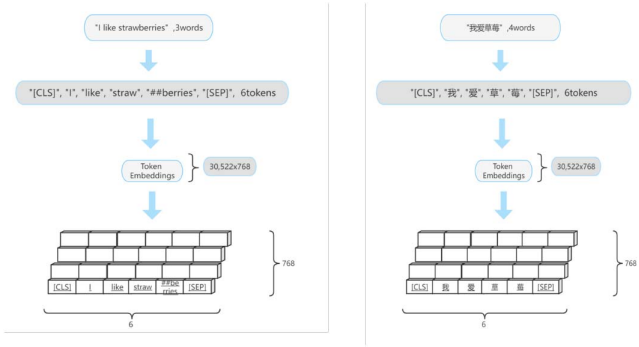


Fig. 3. Chinese–English segmentation differences.

on the training subset of the dataset. Following this, TRALSem vectors the data that has undergone segmentation.

To bolster the model's resilience, adversarial training is integrated. Next, the data is batched (with max_len = 256 and batch size = 16), and each data point is processed sequentially through a transformer encoder comprising 12 layers to generate the vector for classification results.

In the final stage, a layer implementing linear regression is added to convert the classification task to a task based on linear regression.

For the values that are continuously produced at the results, classification is determined by determining the element with the highest value within the output that has been adjusted. To ensure the ability to replicate data associated with the model and to minimize algorithmic variability, RSs are utilized for consistency.

B. Data Encoder With Adversarial Training

The model's input procedure, as depicted in Part A of Fig. 1, is a result of the experiment. As illustrated in Fig. 2, the vector of input is constructed by combining three types of embeddings: token, position, and segment text, all of which are used as input for a specific style.

WordPiece Embeddings: These embeddings represent the vectorized form of individual words. WordPiece, as the name suggests, divides a group of words into a finite set of shared subunits, achieving a balance between word validity and character-level flexibility. By representing words as vectors, this method becomes computationally efficient for machine learning models.

Position Embeddings: These embeddings encode the positional information of words into specific vectors, transforming them into features for the model. Since RNN and LSTM architectures are not used in this context, sequence positional information cannot be captured inherently. However, such positional information is crucial for learning and predictive tasks in downstream models. To address this, Position Embeddings are constructed as part of the Transformer framework, which encodes positional information using predefined rules. In BERT and its variants, the maximum sequence length is 512 tokens. The position embeddings layer in these models is implemented as a lookup table of size (512, 768). The first row of this table corresponds to the positional vector for the first token in the sequence, the second row corresponds to the second token, and so on, covering the entire sequence. For example, in two sentences, "Hello Bob" and "Hi there", the words "Hello" and "Hi" would share identical position embeddings since they are both the first tokens in their respective sequences. Similarly, "Bob" and "there" would have the same position embeddings.

Segment Embeddings: These embeddings are vector representations used to differentiate between two sentences. This is particularly useful in tasks involving asymmetric sentence structures, such as question-answering scenarios.

BERT and similar models combine word embeddings, position embeddings, and segment embeddings as input to the model. Notably, in the English BERT model, vocabulary tokens are further split into finer-grained semantic units using the WordPiece algorithm. As illustrated in Fig. 3, for example, the word "strawberries" is tokenized into "straw" and "##berries". In contrast, for Chinese, input text is not tokenized into words; instead, individual characters serve as the basic units of text representation.

When using the data encoder, we employed three optimized strategies: adding the RS, incorporating adversarial training, and adding a LL. The methods for adversarial training include the fast gradient method (FGM) [30], PGD [31], free large-batch (FreeLB) [32], and YOPO [33]. Robustness is a critical issue in many machine learning tasks. For example, in natural language processing, models can be sensitive to minor textual perturbations, such as spelling errors or synonym substitutions, leading to incorrect predictions. Furthermore, during model training, adversarial examples generated by the FGM can be used alongside the original samples to train the model. This enables the model to learn more robust feature representations, making it better equipped to handle various potential input variations and disturbances in real-world applications. This approach is particularly significant for improving the generalization ability and reliability of models.

For the input text $X(x_i \mid i = 1, 2..., T)$, Z is the associated classification outcome. The text is processed utilizing the model's tokenizer. Following this, we do the data adversarial challenge with the FGM, introduce disturbances to training examples, and augment perturbations in the samples. The specific calculation formula is as follows:

$$g = \nabla_x L(\theta, x, y) \tag{1}$$

where g represents the gradient, $L(\theta, x, y)$ is the loss function with respect to the parameters θ, the input x, and the true label y,

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and ∇x denotes the gradient with respect to the input.

$$r_{\text{adv}} = \varepsilon \bullet g / \| g \|_2. \tag{2}$$

Here, ε is a hyperparameter, typically set to 1.0, and r_{adv} denotes the adversarial perturbation, which is computed by multiplying the gradient by the hyperparameter ε and normalizing by the norm of the gradient g

$$x_{\text{adv}} = x + r_{\text{adv}} \tag{3}$$

where ε represents a hyperparameter, with a default setting of 1.0.

PGD is an optimization algorithm widely used in constrained optimization problems. It is an extension of the gradient descent algorithm, which ensures that the solutions during iterations always satisfy the predefined constraints by introducing a projection operation in the iterative updating process of gradient descent. The algorithm has important applications in many fields such as machine learning, signal processing, control theory, etc., and especially shows unique advantages in dealing with optimization problems with constraints. The calculation formula of PGD is as follows:

$$r_{\text{adv}|t+1} = \alpha g_t / \|g_t\|_2. \tag{4}$$

In this equation, $r_{\text{adv}|t+1}$ represents the adversarial perturbation at step $t+1$, α is the step size parameter, g_t is the gradient at step t, and $g_t/\|g_t\|_2$ denotes the norm of the gradient. This equation computes the adversarial perturbation for the next step by scaling the gradient at step t by the step size α and normalizing it by the L_2 norm of the gradient. where

$$\|r\|_2 \leq \epsilon. \tag{5}$$

FreeLB (free large batch optimization for language understanding) is an optimization algorithm proposed for large-scale language understanding tasks in the field of natural language processing. FreeLB introduces the idea of adversarial training. Adversarial training is commonly used to enhance the robustness of a model by adding small perturbations to the input data so that the model is trained to be robust not only to correctly classify the original data but also to the perturbed data. In FreeLB, for each input sample, multiple adversarial samples are generated and these adversarial samples are used along with the original samples as one large batch to update the model parameters. The calculation formula of FreeLB is as follows:

$$r = \epsilon \cdot \text{sign}(\nabla_x L(\theta, x)) \tag{6}$$

where r is the adversarial perturbation, ϵ is a hyperparameter that controls the magnitude of the perturbation, $\text{sign}(\cdot)$ the sign function, and $\nabla_x L(\theta, x)$ is the gradient of the loss function L with respect to the input x.

YOPO is an innovative optimization algorithm designed for training deep neural networks. It represents a significant enhancement from traditional optimization methods by introducing a novel approach to dealing with the computational challenges posed by DL architectures, particularly in terms of computational cost and training speed. YOPO aims to reduce the number of forward and backward passes required during the training process, thereby offering a more efficient way to train

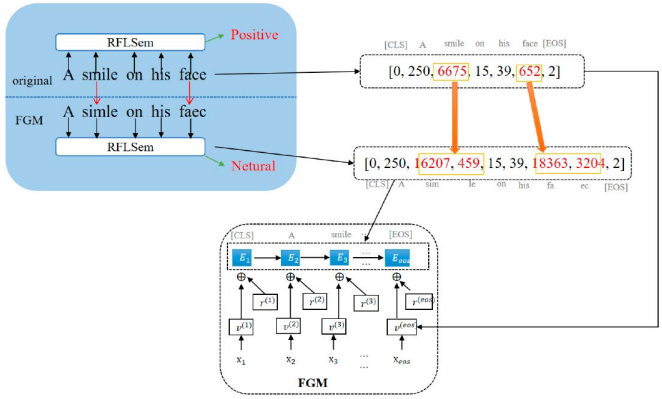


Fig. 4. Example of FGM.

deep neural networks without sacrificing model performance. The calculation formula of YOPO is as follows:

$$r_i^{j,s+1} = r_i^{j,s} + \alpha_1 p \cdot \nabla_{r_i} f_0\left(x_i + r_i^{j,s}, \theta_0\right) \tag{7}$$

where

$$p = \nabla_{g_{\tilde{\theta}}}\left(l\left(g_{\tilde{\theta}}\left(f_0\left(x_i + r_i^{j,0}, \theta_0\right)\right), y_i\right)\right) \cdot \nabla_{f_0}\left(g_{\tilde{\theta}}\left(f_0\left(x_i + r_i^{j,0}, \theta_0\right)\right)\right) \tag{8}$$

where $g_{\tilde{\theta}}$ is an auxiliary function parameterized by θ_0, l is the loss function, and y_i is the true label of the ith sample. Equation (8) computes the necessary value for (7) by calculating the product of the gradients of the loss function concerning both the auxiliary function and the model function. It plays a key intermediate computational role in the YOPO algorithm, providing important information and direction for the perturbation update.

Consider an example to elucidate the function of FGM. As illustrated in Fig. 4, we note that introducing perturbations to the sentence "A smile on his face" yields an adversarial example that maintains semantic coherence with the original. Nonetheless, the sentiment of the original sentence is positive. Following the incorporation of adversarial training techniques, the sentiment shifts to neutral. By doing so, we can create samples that induce errors in the model during training, which in turn amplifies the loss. This approach allows us to subject the model to a greater number of errors, thereby indirectly enhancing the model's robustness.

C. Transformer With RS

The TRALSem utilizes a transformer-based architecture, where the core consists of a stack of twelve layers of transformer encoders. Each transformer encoder is composed of a layer implementing the multihead self-attention mechanism followed by a fully connected feed-forward layer. The transformer encoder integrates three extra components residual connections, layer normalization, and linear transformations. The residual connection involves immediately combining the model's input and output data with the final output, which simplifies the network structure. Layer normalization employs the zero-mean and

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unit variance method to normalize the nodes of a particular neural network layer. To enhance the comprehensive model's expressiveness, a linear transformation is applied, which performs two linear transformations on the semantic representation of each term, ensuring that the resulting vector maintains an equivalent length to the initial vector.

The multihead attention layer processes queries (Q), keys (K), values (V), and other relevant details via parameterized matrices, performing a self-attention operation. This data is subsequently incorporated into a fully connected feed-forward layer. The weight attributed to each value is denoted as the similarity function between the key and its corresponding query. The computational process is as outlined as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_K}}V\right). \tag{9}$$

The multihead self-attention layer maps the Q, K, and V parameters of a single layer utilizes the parameter matrix, and subsequently executes self-attention. The results are then combined and output through a fully connected layer. The computational process is as follows:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, ..., \text{head}_h)W^O \tag{10}$$

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \tag{11}$$

where $W_{\text{i}}^Q \in R^{d_{\text{model}} \times d_Q}$, $W_{\text{i}}^K \in R^{d_{\text{model}} \times d_K}$, $W_{\text{i}}^V \in R^{d_{\text{model}} \times d_V}$, $W_{\text{i}}^O \in R^{d_{\text{model}} \times hd_V}$.

The feedforward fully connected layer module comprises two linear transformation modules, with a ReLU activation function positioned between them. The corresponding mathematical formula is as follows:

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2 \tag{12}$$

where $\text{FFN}(x)$ represents the output of the feed-forward fully connected layer, and x is the input. W_1 and W_2 are the weight matrices for two linear transformations, and b_1 and b_2 are the corresponding bias vectors. The formula first multiplies the input x with the weight matrix W_1 and adds the bias vector b_1, then passes through the ReLU activation function (i.e., takes taking the maximum value of 0 and the result of $xW_1 + b_1$), and finally multiplies it with the weight matrix W_2 and adds the bias vector W_2 to get the final output. This structure is common in feed-forward neural networks, where the introduction of the ReLU activation function adds nonlinearity, allowing the network to learn more complex functional mappings. This enhances the network's expressive and learning capacity, playing a critical role in DL models, and enabling better data fitting and feature extraction.

We utilize the linear congruential generator (LCG) method to establish RSs. This method, referred to as the linear congruential generator, is used to produce pseudorandom numbers. The computational process is as outlined as follows:

$$\text{RandSeed} = (A * \text{RandSeed} + B). \tag{13}$$

The core elements of the LCG are the multiplier A, the increment B, and the modulus M, which are constants specified by the generator. The greatest achievable period for the LCG is determined by the value of M, although it is typically shorter than M. The values of A, B, and M must compile with the following conditions.

1) B and M are coprime, indicating that they share no common factors other than 1.
2) Any prime factor of M must be a divisor of A-1.
3) If M is divisible by 4, then A-1 must also be divisible by 4.
4) M is greater than A and B. A and B are positive integers.

D. Adding LL

The LL is applied to conduct regression calculations, which serve as a form of logical classification. For the numerical values produced at the model's output, classification is achieved by identifying the part with the highest value in the fitted output. This method aids in the extraction of sentiment values for specific sentiment classification tasks, aligning with the requirements of downstream tasks. Additionally, we incorporate the Dropout function to reduce overfitting. The computational process is as outlined as follows:

$$z_i = y_i \bullet w + b. \tag{14}$$

In this context, y_i represents the output. Finally, the classification result can be determined using the following equation:

$$Z_i = \text{softmax}(z_i). \tag{15}$$

The softmax layer is primarily utilized to finalize the sentiment classification. The softmax function converts the raw outputs of the model's final layer (typically a real-valued vector) into a probability distribution, where each element corresponds to a specific class, and the sum of all elements equals 1. This transformation allows the model's output to be interpreted as the probability of the input text belonging to each class. For instance, an output vector [0.2, 0.7, 0.1] indicates that the probability of the input text belonging to the positive class is 0.7, the negative class is 0.2, and the neutral class is 0.1. This probability-based output enhances the interpretability of the model's predictions, enabling users to intuitively understand the model's inclination toward each class.

E. Training

The training dataset encompasses a variety of domains. Furthermore, the input text may consist of several statements or merely a solitary statement. To ensure consistent training, we impose a cap on the input text length, which must not be more than 256 characters. Should this threshold be exceeded, we will segment the data in the preprocessing stage. As training advances and data is backpropagated, we employ the Adam optimization algorithm

$$\theta_{t+1} = (1 - \lambda)\theta_t - \alpha M_t \nabla f_t(\theta_t) \tag{16}$$

where θ_{t+1} represents the parameter value at step $t + 1$, θ is the parameter value at step t, λ is a decay rate parameter, αM_t is the learning rate, M_t is the first-order moment estimate, and $\nabla f_t(\theta_t)$ is the gradient of the objective function f_t with respect to θ_t. Additionally, we introduce a LL to convert the initial

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classification task into a regression task. This approach helps the model better adapt to various types of tasks and data, improving its generalization ability and performance. During training, by limiting the input text length and using efficient optimization algorithms (such as Adam [54]), the training process becomes more efficient and stable, allowing the model to learn better patterns and features from the data, leading to improved performance across various tasks.

Furthermore, we introduce a LL to transform the initial classification task into a regression task. Consequently, we employ root mean square error (RMSE) as the loss function

$$\text{loss}(x_i, y_i) = \sqrt{(x_i - y_i)^2} \tag{17}$$

where x_i denotes the true value, and y_i denotes the predicted value. Through this process, we derive a continuous value from the output once it has passed through the LL. We can then use this value as the sentiment score, which can then be mapped to a distinct category in accordance with the sentiment classification standards of the specific dataset.

During the training process, our framework utilizes training datasets that include both Chinese and English data. For different languages, we employ distinct encoder models (e.g., Google BERT base model (uncased)[1], and Google BERT base model for Chinese[2]), and the pretrained models used in our framework include both Chinese and English versions.

IV. Experiments

A. Research Questions

Our research design was structured to tackle the following six research questions.

1) *RQ1:* What is the effectiveness of TRALSem compared with traditional methods?
2) *RQ2:* What is the effectiveness of TRALSem compared with the pretrained models currently in use?
3) *RQ3:* What is the effectiveness of TRALSem as compared with standard pretrained models post-optimization?
4) *RQ4:* What is the impact of the softmax layer and the LL for TRALSem?
5) *RQ5:* What is the impact of the optimization strategies for TRALSem?
6) *RQ6:* What is the impact of the robustness strategies for TRALSem?

B. Subjects Under Study

For this experiment, four datasets are utilized, as outlined in Table I. They comprise the Chinese dataset (CData)[3], the English dataset (EData)[4], the IMDB dataset[5], and the Stanford Sentiment Treebank (SST)[6]. As shown in Table I, the granularity of labeling is different for the datasets we used, which suggests that the granularity of sentiment categorization during our experiments was also different.

TABLE I
Statistics of the Dataset

Name	Training Set	Testing Set	Categories
CData	36 782	21 376	6
EData	40 000	20 000	2
IMDB	40 000	10 000	2
SST	8454	2200	5

The dataset of Chinese is categorized into love, happiness, shock, anger, fear, and sorrow classes. It includes a subset with 36 782 entries for training and a subset comprising 21 376 entries for testing. The English dataset, sourced from the Kaggle platform, features dialogue attributed to characters, with sentiments classified as either negative (0) or positive (1). This dataset is divided into a set of 40 000 entries for training and a set of 20 000 entries for testing.

The IMDB dataset, which is a collection of movie reviews, encompasses 50 430 entries across 28 columns. The publicly available version of this dataset is currently a binary (positive rating of 1, negative rating of 0) classification dataset. In our study, we concentrated exclusively on the review text and sentiment columns, which led to a training dataset comprising 40 000 entries and a separate test dataset with 10 000 entries.

The SST is a sentiment benchmark dataset. The sentiment scores are mapped to intervals [0, 0.2], (0.2, 0.4], (0.4, 0.6], (0.6, 0.8], (0.8, 1.0], which we translated into five sentiment categories: 0, 1, 2, 3, and 4. The training subset consists of 8454 entries, while the test subset contains 2200 entries.

The data preprocessing primarily focuses on CData. The CData, which includes Chinese script training and testing data, was provided by iQIYI for the 2021 CCF competition[7]. The dataset is mainly sourced from a selection of movie scripts, as well as sentiment annotations from iQIYI's labeling team. In the original data, character names are represented as lower-case letters followed by numbers, and some character names are repeated across different scripts. Therefore, based on the script number and scene, we randomly generated three-character Chinese names to replace the original numbered character names. Since a single sentence may contain different subjects and the context also affects sentiment analysis, we concatenated n sentences from the same scene (with n sentences not exceeding 512 bytes in length) in this experiment. Additionally, we experimented with concatenating n sentences of the same character's context, but the results were not satisfactory. This is likely because the context of a single character over n sentences can span a large range, thus affecting the model's sentiment judgment. For the other datasets, we used publicly available datasets from existing studies and, as with other research, no special processing was applied to these datasets.

[1]https://huggingface.co/google-bert/bert-base-uncased
[2]https://huggingface.co/google-bert/bert-base-chinese
[3]https://www.datafountain.cn/competitions/518
[4]https://www.kaggle.com/datasets/lowoncuties/czech-movie-review-csfd
[5]https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews
[6]https://nlp.stanford.edu/sentiment/code.htm
[7]https://www.datafountain.cn/competitions/518

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TABLE II
COMPARISON WITH TYPICAL SENTIMENT ANALYSIS TECHNIQUES

Data	Approach	Siebert	Bart	J-Hartmann	SNLP	SENT	LTCNN	GRU	RNN	FNN	TRALSem
Movie-ch	Macro-F1	0.5140	0.2900	0.3000	0.2511	0.2302	0.1421	0.1421	0.2402	0.1421	**0.7982**
	Micro-F1	0.7233	0.2905	0.2900	0.2543	0.2318	0.7430	0.6333	0.7431	0.6335	**0.9313**
	RSME	0.7011	0.7433	0.7324	0.7246	0.7433	0.8082	0.8088	0.8081	0.8083	**0.3376**
	Precision	0.5270	0.3417	0.3616	0.2186	0.1715	0.1266	0.1520	0.1521	0.1522	**0.8988**
	Recall	0.5019	0.2501	0.2590	0.3098	0.3690	0.1667	0.1430	0.1433	0.1431	**0.8632**
Movie-en	Macro-F1	**0.9527**	0.3333	0.3105	0.3333	0.6005	0.4808	0.4920	0.4921	0.4806	0.9499
	Micro-F1	**0.9532**	0.3333	0.2797	0.3333	0.6005	0.4808	0.4920	0.4921	0.4806	0.9499
	RSME	**0.1546**	0.3451	0.4031	0.7071	0.6098	0.7078	0.7071	0.7070	0.7080	0.1640
	Precision	**0.9531**	0.2500	0.2250	0.2500	0.6768	0.4989	0.4990	0.5000	0.4986	0.9523
	Recall	0.9523	0.5000	0.5000	0.5000	0.6281	0.4990	0.5001	0.5001	0.4988	**0.9523**
IMDB	Macro-F1	0.9356	0.3000	0.3110	0.3330	0.4702	0.4914	0.5006	0.4973	0.4954	**0.9523**
	Micro-F1	0.9356	0.3325	0.2807	0.3330	0.4702	0.4914	0.5006	0.4973	0.4954	**0.9523**
	RSME	0.1356	0.3446	0.4336	0.7076	0.7026	0.7075	0.7053	0.7076	0.7089	**0.1307**
	Precision	0.9364	0.2496	0.2257	0.2496	0.5093	0.4992	0.5027	0.4992	0.4974	**0.9523**
	Recall	0.9349	0.5000	0.5000	0.5000	0.5068	0.4993	0.5026	0.4993	0.4974	**0.9523**
SST	Macro-F1	0.3424	0.2697	0.2743	0.3010	0.2833	0.1676	0.1720	0.1275	0.1721	**0.4933**
	Micro-F1	0.3514	0.2874	0.2744	0.3140	0.2789	0.2034	0.1981	0.1666	0.1996	**0.5051**
	RSME	0.6647	0.8843	0.8735	**0.7056**	0.7135	0.8250	0.9300	0.8666	0.7488	0.7419
	Precision	0.3433	0.4047	0.4104	0.3329	0.3001	0.1429	0.1579	0.0971	0.1979	**0.5266**
	Recall	0.3529	0.4253	0.4137	0.3402	0.2998	0.2061	0.1967	0.1872	0.1984	**0.5647**

Note: The bolded entries indicate the groups with the best results under that set of experiments.

C. Variables

1) Independent Variables – Approaches Being Evaluated: The variable under examination in our research focuses on the methodologies that are to be assessed and compared. Our introduced method was inherently chosen for inclusion in the study. As previously mentioned, the research employs two kinds of connected graphs—those centered on nouns and those centered on verbs. Consequently, two variations of our method will be subject to evaluation in the experiments: TRALSem-Noun and TRALSem-Verb.

To facilitate a comprehensive comparison, we selected 17 baseline techniques (11 for RQ1 and 6 for RQ2), as detailed below. For RQ1, we opted for 11 conventional sentiment analysis approaches: Stanford CoreNLP (SNLP) [55], Sentistrength (SENT) [56], GRU [57], CNN [24], FNN [58], and LSTM+TextCNN (LTCNN) [27].

In the context of RQ2, we assess a range of standard pretrained models: BERT [59], Roberta [60], Albert [61], XLNet [62], ERNIE [63], and GPT [64].

In the context of RQ3, we utilize our model optimization technique to assess the performance.

In RQ4, we evaluate the performance of combining the softmax activation function with a LL and using it separately.

For RQ5, The three optimization strategies (OS) are individually applied to the model for refinement.

For RQ6, we compare four attack strategies and evaluate the robustness of the model under the strategies.

2) Dependent Variables – Evaluation Metrics and Parameter Settings: For the evaluation of RQ1 through RQ6, we selected the metrics Precision, Recall, RSME [65], macro-F1 [66], and micro-F1 [67] to assess the accuracy of our predictions on the test sets. RMSE serves as a measure of the dispersion of these residuals.

In RQ1 and RQ2, we conducted a comparative analysis with a consistent set of epoch and learning rate parameters. The chosen epoch is 4, and the learning rate is set to 2×10^{-5}. All other hyperparameters of the models were set to the default values provided for the models on HuggingFace[8]. For example, num_hidden_layers is 12 layers, num_attention_heads is 12, and intermediate_size is 3072. The training was conducted on a server equipped with an NVIDIA 3090 GPU.

In RQ3, we also compared the training time of our model method with the existing methods. The main idea is to record the timestamps before and after the training of the model and calculate the difference. The equation is as follows:

$$\text{Time} = \text{Time}_{\text{begin}} - \text{Time}_{\text{end}}. \tag{18}$$

V. RESULTS AND ANALYSIS

A. Answer to RQ1

We reproduced 11 typical sentiment analysis techniques based on their original algorithms. The comparison results with our method are summarized in Table II. TRALSem outperforms the existing methods across datasets.

To quantify the extent of performance enhancement, we analyzed the results. As seen in Table II, the effectiveness of the TRALSem sentiment analysis approach is notably superior to those typical approaches. Stanford CoreNLP constructs sentence representations based on sentence structure; however, it lacks integration of contextual knowledge. SentiStrength assesses emotions at the sentence level by giving points to positive and negative words in isolation and then aggregating these points. This approach overlooks word order and word relationships. GRU, CNN, FNN, and LSTM + TextCNN utilize the Bert

[8]https://huggingface.co/

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TABLE III
COMPARISON WITH TYPICAL PRETRAINED MODELS

Data	Approach	BERT	RoBERTa	ALBERT	XLNet	ERNIE	GPT	RFLSem	TRALSem
Movie-ch	Macro-F1	0.7641	0.7668	0.7533	0.7610	0.7644	0.6806	0.7758	**0.7982**
	Micro-F1	0.9123	0.9210	0.9110	0.9092	0.9100	0.8687	0.9247	**0.9313**
	RSME	0.4026	0.4193	0.4008	0.4102	0.3975	0.4506	0.3553	**0.3376**
	Precision	0.8130	0.8746	0.8004	0.8655	0.8337	0.8034	0.8821	**0.8988**
	Recall	0.8201	0.8700	0.8155	0.8532	0.8320	0.8142	**0.8745**	0.8632
Movie-en	Macro-F1	0.9375	0.9515	0.9305	0.9508	0.9211	0.8677	**0.9620**	0.9499
	Micro-F1	0.9375	0.9515	0.9305	0.9508	0.9211	0.8677	**0.9620**	0.9499
	RSME	0.1455	**0.1331**	0.2064	0.1361	0.2238	0.4015	0.1380	0.1640
	Precision	0.9425	0.9484	0.9300	0.9475	0.9213	0.8550	**0.9648**	0.9523
	Recall	0.9422	0.9483	0.9300	0.9469	0.9245	0.8810	**0.9549**	0.9523
IMDB	Macro-F1	0.9121	0.9108	0.9108	0.9322	0.9209	0.8002	0.9437	**0.9523**
	Micro-F1	0.9121	0.9108	0.9108	0.9322	0.9209	0.8002	0.9437	**0.9523**
	RSME	0.3267	0.2346	0.4113	0.2675	0.3841	0.4769	**0.1300**	0.1307
	Precision	0.9154	0.9122	0.9122	0.9322	0.8011	0.8019	0.9438	**0.9523**
	Recall	0.9122	0.9108	0.9108	0.9322	0.8045	0.7986	0.9437	**0.9523**
SST	Macro-F1	0.4951	0.5360	0.4493	0.5077	0.5124	0.4123	**0.5457**	0.4933
	Micro-F1	0.5087	**0.5568**	0.4627	0.5127	0.5200	0.4211	0.5512	0.5051
	RSME	0.7436	0.7391	0.8970	0.7647	**0.7344**	0.7955	0.7567	0.7419
	Precision	0.5212	0.5380	0.4751	0.5208	0.5289	0.5016	**0.5435**	0.5266
	Recall	0.4933	0.5069	0.4542	0.5135	0.5011	0.5108	0.5530	**0.5547**

Note: The bolded entries indicate the groups with the best results under that set of experiments.

model for tokenization and integrate neural networks as additional layers on top, but these networks depend on extensive training data and require self-training, resulting in evaluation index scores that are not as favorable as those of pretrained models.

In conclusion, our method substantially enhances the effectiveness of sentiment analysis tasks in comparison to traditional machine learning and DL-based approaches.

B. Answer to RQ2

The outcomes for RQ2 are presented in Table III. As can be seen in Table III, our approach generally demonstrated the highest effectiveness across most of the datasets.

The TRALSem model's architecture is akin to that of BERT, Roberta, and Albert. However, unlike BERT, Roberta, and Albert, TRALSem incorporates a LL to convert the classification task into a logical classification problem. Moreover, during the training process, TRALSem introduces RSs and adversarial training techniques to bolster the integrity of the data and minimize the effects of data perturbations on the model's training process.

The XLNet model is well-suited for processing extended textual content. If the dataset is limited and the sentences within the sentiment analysis context are brief, the full potential of the XLNet model may not be completely realized. The GPT employs the transformer's decoder to mask underlying material. Consequently, for prediction purposes, the GPT primarily takes into account one-directional information, which can diminish its predictive effectiveness.

The ERNIE model facilitates the learning of complete concept semantic representations by masking semantic units such as words and entities. Unlike BERT, which learns from the original language signals, ERNIE directly models prior semantic knowledge units, thereby enhancing the model's semantic representation capabilities. ERNIE's training corpus incorporates data from multiple sources with knowledge content. However, ERNIE employs knowledge entities as the masking granularity, effectively introducing knowledge priors. This approach has limitations in domains with partially open knowledge entities.

The effect of TRALSem is also improved compared with our previously published RFLSem [1], because we have upgraded the module of robustness from the original one, and chose a better data perturbation model to improve the effect of the whole framework.

In conclusion, it is evident that our methodology outperforms the six classical pre-trained models across all four datasets.

C. Answer to RQ3

The outcomes of the experiments addressing RQ3 are displayed in Table IV. From Table IV, it is evident that TRALSem exhibits the most favorable outcomes. This superiority is attributed to the context-dependent nature of sentiment analysis, which necessitates a model that can capture nuanced semantic relationships. ERNIE enhances the model's generalizability by integrating masks and enabling the learning of knowledge dependencies and longer-term semantic dependencies. Conversely, the improvements made to the Albert model are primarily focused on natural sentence split (NSP) tasks, which are less relevant to sentiment analysis tasks.

We note that the general performance indicator of the English dataset is superior to that of the Chinese dataset. This is mainly attributed to the English vocabulary being extensively trained on a comprehensive corpus. However, TRALSem does not separate Chinese input into individual characters, treating each

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TABLE IV
COMPARISON WITH THE OPTIMIZED PRETRAINED MODELS

Data	Approach	BERT	ALBERT	XLNet	ERNIE	GPT	RFLSem	TRALSem
Movie-ch	Macro-F1	0.7755	0.7587	0.7623	0.7674	0.7541	0.7758	**0.7982**
	Micro-F1	0.9247	0.9180	0.9210	0.9215	0.8791	0.9247	**0.9313**
	RSME	0.3872	0.3945	0.3839	0.3698	0.4002	0.3553	**0.3376**
	Precision	0.8744	0.8634	0.8709	0.8655	0.8554	0.8821	**0.8988**
	Recall	0.8732	0.8501	0.8712	0.8439	0.8530	**0.8745**	0.8632
Movie-en	Macro-F1	0.9394	0.9349	0.9523	0.9201	0.8006	**0.9620**	0.9499
	Micro-F1	0.9393	0.9349	0.9523	0.9201	0.9015	**0.9620**	0.9499
	RSME	0.1459	0.2171	**0.1356**	0.2310	0.3901	0.1380	0.1640
	Precision	0.9531	0.9448	0.9527	0.9438	0.8590	**0.9648**	0.9523
	Recall	0.9514	0.9502	0.9533	0.9367	0.8921	**0.9549**	0.9523
IMDB	Macro-F1	0.9190	0.9046	0,9303	0.9289	0.8853	0.9437	**0.9523**
	Micro-F1	0.9190	0.9046	0.9303	0.9289	0.8853	0.9437	**0.9523**
	RSME	0.1386	0.1329	**0.1262**	0.1400	0.3901	0.1300	0.1307
	Precision	0.9200	0.9062	0.9306	0.9114	0.8590	0.9438	**0.9523**
	Recall	0.9198	0.9047	0.9303	0.9120	0.8921	0.9437	**0.9523**
SST	Macro-F1	0.4992	0.4634	0.5288	0.5310	0.5273	**0.5457**	0.4933
	Micro-F1	0.5089	0.4786	0.5310	0.5324	0.5301	**0.5512**	0.5051
	RSME	0.7438	0.7733	0.7678	0.7723	0.7810	0.7900	**0.7419**
	Precision	0.5139	0.4787	0.5242	0.5279	0.4689	**0.5435**	0.5266
	Recall	0.4948	0.4598	0.5493	0.5489	0.5105	0.5530	**0.5647**

Note: The bolded entries indicate the groups with the best results under that set of experiments.

word as a standalone unit. Additionally, the selected English dataset primarily consists of short statements and sentiment analysis focused on particular aspects, where a single sentence serves as the learning unit, and the records are not emotionally connected or impactful.

We compare the effect of the basic model in terms of the time dimension after using our lightweight framework, as shown in Fig. 5. The base time in the figure denotes the time for baseline model training, and the pro time denotes the time for training after adding our TRALSem framework to the baseline pretrained model. We can see that the training time of the model is significantly optimized after using the framework, which further demonstrates the efficiency of our framework. As we can see from Fig. 5, the training time of our method improves on average by more than 90% compared with the baseline model, which is a good indication of the purpose of our lightweight model. TRALSem significantly reduces the training time of the model on downstream tasks.

In Fig. 6, we show the loss function of the basic model and the training process of our TRALSem. The loss of Pro loss shows better stability and optimization effect without significantly high fluctuation, indicating that the improved experimental optimization converges faster and that our TRALSem framework is more effective. The fluctuation of Base loss is more significant, indicating that the base experimental model may have gradient fluctuation or slower convergence during the training process.

We also explored the case of a smaller scale of training samples, we screened the CData and selected 10 000 data for training. The experimental results indicate that the training time for training samples has been reduced. Despite this reduction in training time, compared with the original model, various parameter indicators of the training results have still been improved.

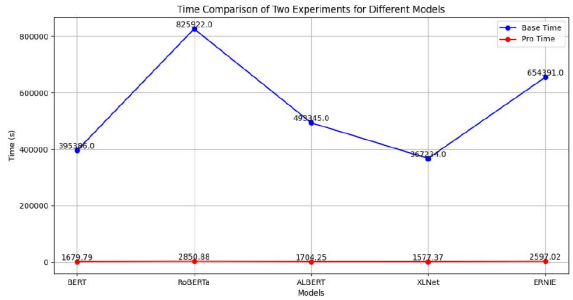


Fig. 5. Time comparison of two experiments for different models.

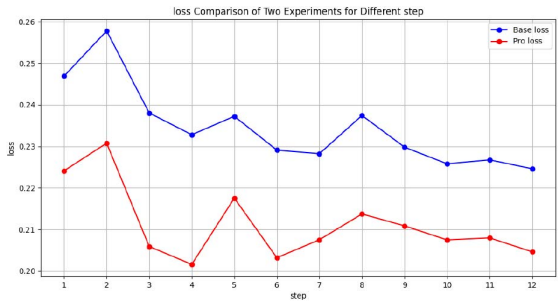


Fig. 6. Loss comparison of two experiments for different step.

However, when compared with the results obtained from large-scale training data, there is still a decline. This phenomenon clearly demonstrates that the model yields better results when more training data are available.

To summarize, TRALSem outperforms the six optimized pretrained models in terms of performance.

D. Answer to RQ4

When our downstream tasks are associated with emotionally sensitive work, such as recommender systems, the Softmax

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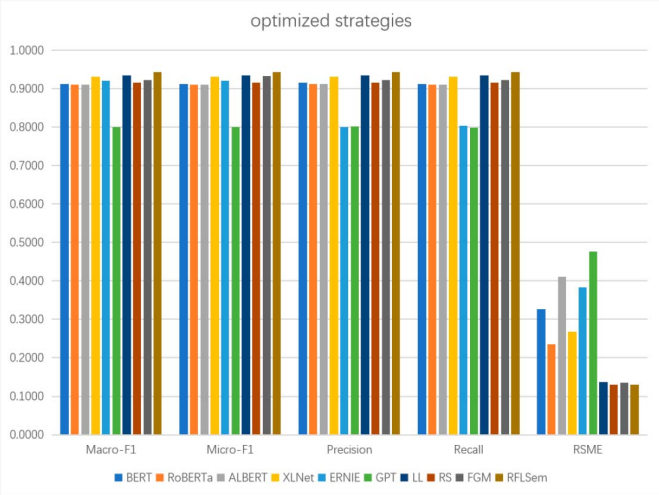


Fig. 7. Comparison results of different optimized strategies.

layer's mapping to interval probabilities may compromise the accuracy of the system. By incorporating a LL, we observe improved results on three datasets. This combined approach not only addresses the overfitting tendency of the LL but also resolves the Softmax output's limitation, where a sample can only be assigned to one category. Softmax is less suitable for tasks that require handling multiple emotional dimensions. In the context of recommender systems, which often involve the fusion of multiple emotions, the combination of the two methods effectively reflects the multidimensional emotional aspects. This is evident in the two multiclassification datasets, Movie-ch and SST.

In conclusion, TRALSem with the combination of softmax and LLs achieves better results than the variants with softmax and LLs alone.

E. Answer to RQ5

The outcomes of the experiments addressing RQ5 are shown in Fig. 7.

We find that the application of three optimization techniques outperforms the results obtained from six pretrained models. The inclusion of a linear regression layer and FGM adversarial training has a modest positive impact on the model's effectiveness. The addition of the linear regression layer transforms the classification task into a linear regression task, which entails categorizing the logical classification and the continuous output value by identifying the maximum element within the fitted output. This adjustment leads to a slight increase in accuracy compared with directly using classification result vectors. FGM adversarial training, which involves introducing noise during training, helps in regularizing the model's parameters, thereby enhancing its robustness and generalization capabilities, and improving its performance in practical applications. The optimization effect is not strikingly pronounced with RS. This is because the role of RS is not to optimize or enhance the model's evaluation metrics but rather to ensure reproducibility by randomizing the initial parameters each time the model is initiated.

To summarize, all three optimization strategies have a positive impact on the performance of TRALSem.

F. RQ6 – Ablation Study About Robustness Strategies

The outcomes of the experiments addressing RQ6 are given in Table VI, which shows the results of the verification of the robustness strategy involved in the TRALSem model on the four datasets.

We can observe the result of four robustness strategies. The FreeLB and YOPO algorithms worked well on the four datasets. FreeLB and YOPO are more robust against attacks than FGM and PGD but require more training time and computing resources. FGM determines which dimensions are aggressive to text through the sign of gradient values, and then perturbates these dimensions to generate adversarial samples, which ultimately helps the model improve robustness. The PGD algorithm is based on FGM for multiple iterations to find the optimal perturbation. FreeLB improves the resilience and applicability of the model performance of the model by combining label smoothing and adversarial training. At the same time, FreeLB uses a repeated-measure learning method to reduce the influence of label noise on the model. YOPO realizes text adversarial training through adaptive gradient clipping and backpropagation, which improves the robustness and generalization performance of the model and avoids the problem of excessive perturbation to a certain extent.

Compared with their respective use cases, FreeLB is suitable for dealing with tag noise, while YOPO is designed to balance robustness and accuracy by reducing the number of adversarial samples. Therefore, FreeLB and YOPO are more applicable to our text sentiment classification task.

To summarize, TRALSem performs better with the four robustness strategies, and FreeLB and YOPO perform better than FGM and PGD.

G. Discussion

The Need for a Robust Model of Sentiment Analysis. We found that TRALSem has a better sentiment analysis performance of robustness. However, in the early stage of our experiment, we randomly selected training data and test data to train and predict the model. It was found that differences exist in the results obtained for different randomly selected data sets. For example, we randomly extracted 2000 pieces in the training set and the test set to form a new dataset, and then experiment under the same model led to the Marco-F1 index of 0.9394 and the Micro-F1 index of 0.9393. However, some of the randomly selected data sets performed well, the Marco-F1 index was 0.9805, and the Micro-F1 index was 0.9805.

The Granularity of Sentiment Analysis has a Greater Impact on Classification. We also found that TRALSem has better sentiment analysis performance for sentiment analysis with finer granularity. This is mainly because a difference in sentiment analysis granularity normally exists. Some sentiment analysis methods are categorical, such as binary classification, while other methods give emotional scores. The finer the granularity

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TABLE V
COMPARISON RESULTS OF SOFTMAX LAYER AND LL

Data	Approach	Macro-F1	Micro-F1	RSME	Precision	Recall
Movie-ch	softmax	0.7780	0.9301	0.3527	0.8845	0.8790
	Linear	0.7758	0.9247	0.3553	0.8821	0.8745
	S + L	**0.7802**	**0.9314**	**0.3511**	**0.8894**	**0.8821**
Movie-en	softmax	**0.9557**	**0.9557**	**0.1546**	**0.9560**	**0.9557**
	Linear	0.9536	0.9536	0.1607	0.9541	0.9536
	S + L	0.9493	0.9493	0.1664	0.9513	0.9493
IMDB	softmax	0.9324	0.9324	0.1356	0.9339	0.9325
	Linear	0.9388	0.9388	0.1340	0.9389	0.9388
	S + L	**0.9400**	**0.9400**	**0.1321**	**0.9403**	**0.9403**
SST	softmax	0.4794	0.5004	**0.7465**	0.5041	0.4756
	Linear	0.4582	0.4771	0.7890	0.5029	0.4697
	S + L	**0.4849**	**0.5042**	0.7567	**0.5259**	**0.5244**

Note: The bolded entries indicate the groups with the best results under that set of experiments.

TABLE VI
COMPARISON WITH THE ROBUSTNESS STRATEGIES

Data	Approach	FGM	PGD	FreeLB	YOPO
Movie_ch	Macro-F1	0.7802	0.7900	0.7980	**0.7982**
	Micro-F1	0.9314	0.9303	**0.9342**	0.9313
	RSME	0.3511	0.3523	0.3499	**0.3376**
	Precision	0.8894	0.8905	0.8956	**0.8988**
	Recall	0.8821	0.8846	**0.8874**	0.8632
Movie_en	Macro-F1	0.9493	0.9499	0.9489	**0.9499**
	Micro-F1	0.9493	0.9499	0.9489	**0.9499**
	RSME	0.1664	**0.1632**	0.1721	0.1640
	Precision	0.9513	**0.9524**	0.9510	0.9523
	Recall	0.9493	0.9520	0.9516	**0.9523**
IMDB	Macro-F1	0.9400	0.9403	0.9412	**0.9523**
	Micro-F1	0.9400	0.9403	0.9412	**0.9523**
	RSME	0.1321	0.1340	0.1322	**0.1307**
	Precision	0.9403	0.9490	0.9521	**0.9523**
	Recall	0.9403	0.9400	0.9523	**0.9523**
SST	Macro-F1	0.4849	0.4930	**0.4957**	0.4933
	Micro-F1	0.5042	0.4998	0.5001	0.5051
	RSME	0.7567	0.7478	**0.7322**	0.7419
	Precision	0.5259	0.5248	0.5221	**0.5266**
	Recall	0.5244	0.5255	0.5601	**0.5647**

Note: The bolded entries indicate the groups with the best results under that set of experiments.

of sentiment analysis, the effectiveness of the sentiment analysis approaches will decline to a limited degree.

The Segmentation Threshold is of Significant Importance in the Field of Sentiment Analysis. In this study, we choose 256 as the segmentation threshold mainly due to the following considerations.

1) *Optimization of Computational Resources:* Processing longer text sequences requires more computational resources and time. For shorter texts, directly inputting them into the model does not impose significant computational burdens and preserves the integrity and semantic coherence of the text. However, when the text exceeds a certain length (e.g., 256 tokens), segmenting the data implies dividing a large task into smaller, more manageable subtasks. This segmentation improves overall computational efficiency, enabling the model to handle more data with limited resources.
2) *Preserving Local Semantic Integrity:* By analyzing the dataset's length, we observe that within a certain length range (not exceeding 256 tokens), the local semantics and contextual information of the text can be captured relatively intact by the model. Segmenting shorter texts might compromise this local semantic integrity, making it difficult for the model to understand the text's meaning accurately. For example, a brief sentence may be semantically cohesive, and segmenting it could cause the model to lose crucial semantic relationships and contextual clues.
3) *Avoiding Excessive Segmentation and Information Loss:* While longer texts require segmentation, setting the segmentation threshold too low (e.g., much less than 256 tokens) may lead to excessive fragmentation, causing each segment's semantics to become overly fragmented. This makes it challenging for the model to reconstruct the original text's full meaning and logical connections. Setting the threshold at 256 tokens provides a good balance between handling long texts and maintaining the integrity of semantic information, ensuring that the model can still extract sufficient useful information for accurate encoding and learning from the segmented data.
4) *Batch Processing Consistency:* During model training, data is typically organized into batches for processing. If the length of the texts varies significantly, excessive padding within batches may occur, wasting computational resources and potentially affecting the training performance. By segmenting texts longer than 256 tokens, most texts will fall within a concentrated length range, facilitating batch processing and improving training efficiency and stability. For instance, if all texts in a batch are around 256 tokens, only minimal padding is needed. However, if there is significant length disparity, a lot of short texts may require substantial padding or long texts may require complex processing, increasing training complexity and computational cost.

Comparison With Another Relevant Typical Method. To evaluate the performance of our proposed method more comprehensively, we conducted further experiments to compare

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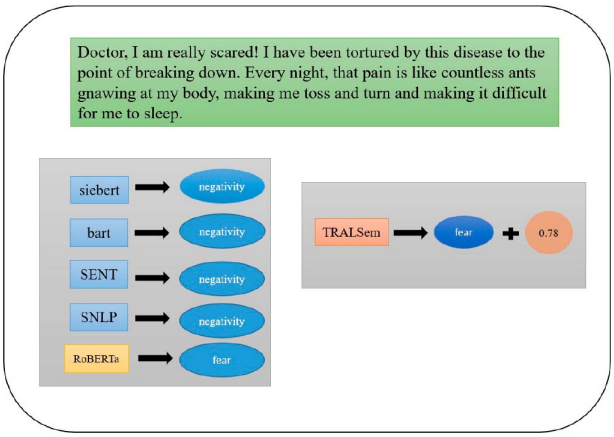


Fig. 8. Examples of classification results for different methods.

TRALSem against the sparse manifold sentiment representation (SMSR) method [68], which also shows high effectiveness in sentiment analysis. Since the original study was not associated with an open-source repository, we strictly reproduced their experiments on the IMDB dataset according to the description given in the original study [68]. In addition, as no technical details were provided on how to implement SMSR on Chinese datasets, we could only conduct experiments on the English dataset. Our experimental results show that the SMSR method achieved a Precision of 0.9361, a recall of 0.9330, and an F1 of 0.9346 on the English IMDB datasets we used in our study. The comparison shows that our TRALSem method outperforms SMSR on all these metrics. Moreover, our method is optimized specifically for noisy data processing, which significantly improves the method's robustness and further expands its potential for application in complex real-world scenarios.

Concrete Examples in Practical Use. To further illustrate the present work, let us look at a real-world example, as shown in Fig. 8: "Doctor, I'm really scared! This disease has tortured me to the point of breaking down. Every night, that pain is like a million ants gnawing at my body, making me toss and turn and making it hard to sleep." On Siebert, Bart, SENT, and SNLP, because of the restricted classification, we can only get classification results of negativity, and on pretrained models such as RoBERTa, because of the free definition of classifiers, we can score FEAR, but none of them have sentiment polarity scores; by contrast, using our TRALSem framework, not only can we get finer classification results, but we can also obtain sentiment scores, which facilitates downstream work.

H. Threats to Validity

Our investigation encounters three principal challenges to its validity. The internal validity is compromised by the inherent limitations of our implementations. Although we have rigorously followed the methodologies outlined in the original studies to replicate baseline techniques, discrepancies in the results may still arise. Moreover, the external validity is significantly threatened by the restricted scope of our datasets, which are confined to Chinese and English. This limitation could potentially undermine the precision and statistical robustness of our experimental evaluations. Finally, while we have undertaken a series of ablation studies to evaluate the efficacy of TRALSem, a more comprehensive experimental scrutiny is imperative to attain a holistic understanding of the proposed technique's overall performance.

VI. Related Work

A. Traditional Approaches Based on Sentiment Dictionary

Traditional sentiment analysis methods that utilize sentiment dictionaries mimic human memory and judgment in a basic form. The goal is to establish a foundational vocabulary and memory, such as recognizing negative words such as "dislike" and "dismiss" and positive words such as "favor" and "appreciate", to create a rudimentary mental lexicon. Subsequently, the input sentences are segmented and compared with the vocabulary stored in memory, enabling the evaluation of emotions based on part-of-speech analysis. This process involves preprocessing, word segmentation, training of emotional vocabulary, and sentiment assessment.

Several studies have investigated the emotional classification capabilities of traditional methods employing sentiment dictionaries. For example, Subasic et al. [69] developed a method for determining the emotional orientation of words by employing artificially constructed emotional polarity lexicons. Kennedy and Inkpen [70] introduced a method that takes into account the contextual emotional polarity of each word. Yan et al. [71] created and utilized a set of Weibo emotional words to augment the basic emotional lexicon. They then applied the expanded lexicon to analyze online comment text and validated the effectiveness of sentiment analysis techniques using the microblog lexicon through their experimental results.

However, traditional sentiment analysis methods that depend on sentiment dictionaries and rule-based evaluation are resource-intensive, requiring substantial human and material resources, including the need for artificial design and prior knowledge. Additionally, the text sentiment analysis approach that utilizes a sentiment dictionary and manually crafted rules exhibits reduced scalability. Since these traditional methods do not necessitate manual tagging, they are relatively easy to understand and implement. However, this approach is heavily reliant on the quality of the dictionary, and the rules are entirely human-derived, limiting the range of content it can handle. The absence of dictionary entries for certain topics makes them more difficult to process, which also impedes research in the field of traditional classification.

B. ML-Based Methods

Numerous researchers have introduced machine learning (ML)-based methods to sentiment analysis. For instance, Bandhakavi et al. [72] introduced the unigram blended model, which broadened the scope of sentiment dictionaries to improve the efficiency of sentiment classification feature extraction. Wawre et al. [73] evaluated and compared two machine learning techniques, support vector machines (SVM) and Naive Bayes (NB), finding that NB generally outperforms other methods in terms

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of classification accuracy when dealing with large-scale training datasets. Huq et al. [23] utilized K-nearest neighbor (KNN) and support vector machine (SVM) algorithms to determine the emotional orientation of Twitter content, with KNN's ability to classify based on text Euclidean distance, proving more effective than SVM's in the classification of positive and negative emotions due to the latter's difficulty in determining the optimal hyperplane. Kaur et al. [74] developed an improved rule-based support vector machine with a global vector (RCNN-GloVe) to address the challenge of accurately detecting emotions within sentences, focusing on abstract nouns and adjectives and demonstrating its efficacy through experimentation. Jin et al. [75] integrated text vectorization techniques with machine classification models to bolster classification capabilities. Farkhund et al. [76] proposed an integrated framework that bridged the gap between dictionary-based and machine-learning approaches, aiming to enhance accuracy and scalability.

C. DL-Based Methods

DL has significantly enhanced the acquisition of profound semantic information within emotions. Socher et al. [77] introduced a model for sentiment analysis of film reviews, which outperformed traditional methods. In recent years, techniques such as CNN, RNN, and LSTM have been increasingly applied to sentiment analysis, yielding impressive results.

Bengio et al. [78] were the first to propose a sentiment analysis method using a single-neuron network. In this model, the neurons of the input layer represent features, the number of layers in the hidden layer is artificially determined, and the number of output layers corresponds to the final classification results.

Currently, various pretraining techniques are widely employed, with pretraining involving the training of a model on an existing dataset. Guo et al. [79] presented a dataset collected from Sina Weibo and approached depression detection as a binary classification problem.

Dai et al. [80] introduced semisupervised sequence learning, an advanced recurrent neural network technique rooted in language models and employing sequence self-coding. Ramachandran et al. [81] expanded upon this work, suggesting that the accuracy of the Seq2Seq [82] model can be enhanced through pretraining. This approach illustrates the effectiveness of pretraining for the Seq2Seq model, fine-tuning a small subset of monitored data under various unsupervised conditions. ELMo [83] introduced a text-based context fusion method, demonstrating positive results in various typical tasks. Recognizing the limitations of static word vectors in Word2Vec and GloVe, which do not account for contextual relationships, ELMo [83] and the BERT model [47] by Deflin et al. address this issue by providing dynamic word vectors. The GPT (generative pretraining) model [84] by Radford et al., based on the Transformer architecture, starts with unsupervised pre-training on a large corpus and then fine-tunes the model for specific tasks. Pota et al. [85] proposed a sentiment analysis method for Twitter, accounting for the noise from emojis and hashtags. They also analyzed tweets in both English and Italian using the BERT model. Xu et al. [86] utilized attention mechanisms at both the word and sentence levels, enabling the learning of insights from each word and sentence.

Nugroho et al. [87] developed a transfer learning model based on BERT, which was specifically designed for analyzing Indonesian user reviews from the top ten apps on Google Play in 2020. They refined the model's hyperparameters to enhance its performance. The effectiveness of the model was assessed using two distinct approaches to label the training data.

Zhang et al. [88] conducted an analysis of sentiment in online course reviews, recognizing the value of accurate sentiment analysis in understanding learner emotions and enhancing course quality. They introduced a fine-tuned bidirectional encoder representation based on the BERT model for targeted sentiment analysis in the context of course reviews.

Sarangi et al. [89] proposed a BERT-based encoder that combines syntax and pretrained embeddings with BERT embeddings through a tandem approach. This method aids in the sentiment analysis of structured text. Wu et al. [90] advanced the field by introducing aspect-based sentiment analysis (ABSA) and target-based sentiment analysis (TABSA), which allow for more sophisticated sentiment interpretations from a single text by utilizing context. They also developed a context-guided BERT model (CG-BERT) to boost sentiment analysis performance by integrating context into the self-attention mechanism.

D. Challenges of Existing Methods

The current research on sentiment analysis is confronted with several challenges. For example, most existing sentiment analysis methods heavily rely on high-quality sentiment dictionaries, which may not cover the dynamic nature of human languages. Scalability issues are prevalent in rule-based and dictionary-based methods, making them less efficient for large datasets or new domains. Both machine learning and DL methods can struggle with contextual understanding, particularly with sarcasm or implied sentiment. DL models are resource-intensive, requiring significant computational power and large labeled datasets. Generalization across different domains remains a challenge, as does handling multilingual and diverse data. Advanced models often lack interpretability, and severe concerns exist about data bias and ethical considerations in the use of sentiment analysis.

VII. Conclusion

In this study, we presented a novel model named TRALSem, which is tailored for sentiment analysis across Chinese and English datasets. A specialized tool was developed to implement the TRALSem model. Through our experiments, we demonstrated that TRALSem has superior performance compared with state-of-the-art techniques. Furthermore, we conducted ablation studies to assess the influence of parameter configurations and optimization strategies.

Regarding future work, further exploration in the following key areas could enhance TRALSem's capabilities and broaden its applications.

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First, investigating the performance of TRALSem in diverse languages and domains beyond English and Chinese would significantly increase its practical value and generalizability. This would involve evaluating its effectiveness on social media, news articles, product reviews, and other relevant contexts.

Second, addressing the challenges of complex sentiment expressions such as sarcasm and irony is crucial for improving the model's accuracy and robustness. This could involve incorporating advanced mechanisms into the model architecture or training on datasets specifically designed to capture these nuances.

Finally, exploring fine-tuning techniques for TRALSem could allow for more efficient adaptation to new tasks and datasets with limited training data. Additionally, focusing on interpretability and explainability would enhance trust and understanding of the model's predictions, while addressing ethical considerations would ensure fairness and mitigate potential biases.

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